

DS-Based Surface Polishing in Simulation: A MuJoCo Reproduction of Passive Impedance Control on a 3/8-Sphere Surface

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Abstract—We present a Python/MuJoCo simulation that reproduces the core ideas of dynamical-systems (DS)-based contact tasks, originally implemented in C++/ROS by Amanhoud *et al.* [1] on a KUKA LWR IV+ manipulator. The target scenario is a Franka Emika Panda robot polishing the outer surface of a 3/8-sphere ($R = 0.15$ m) while maintaining a desired contact force of 5 N. The control architecture combines a DS velocity field that blends a reaching component with a tangential circular limit-cycle, and a passive DS impedance controller (without tank energy) that maps the desired EE velocity to joint torques via $\tau = J^T D(\mathbf{v}_d - \mathbf{v}_{ee}) + \mathbf{g}(\mathbf{q})$. Simulation results demonstrate successful approach, stable surface following, and a repeatable circular polishing trajectory over the 3/8-sphere surface. We discuss the design choices, performance metrics, and the gap between the idealized torque-controlled hardware and the simulated dynamics.

Index Terms—dynamical systems, impedance control, surface polishing, contact tasks, MuJoCo, Franka Panda

I. INTRODUCTION

Physical interaction between robots and their environment remains one of the central challenges in manipulation. Unlike free-space motion, contact tasks require simultaneous control of *position* (staying on the surface) and *force* (applying the right pressure). Classical impedance control [2] achieves a trade-off between the two by defining a target mechanical impedance for the end-effector. However, the desired motion trajectory must still be specified separately.

Dynamical Systems (DS) offer an elegant alternative: the robot's desired velocity is encoded as a globally stable vector field rather than a pre-computed path [3]. When combined with impedance control, DS-based methods can handle variable contact geometry and unexpected perturbations without replanning [1].

Amanhoud *et al.* [1] implemented such a system on a KUKA LWR IV+ robot for circular surface polishing. Their approach uses a passive DS impedance controller augmented with a tank-energy mechanism to guarantee passivity during all phases. In this project, we reproduce the core concepts in simulation—dropping the tank-energy term for simplicity—and

deploy the resulting controller on a Franka Emika Panda robot in MuJoCo [4], polishing the outer surface of a 3/8-sphere.

The main contributions of this work are:

- A faithful Python implementation of the polishing DS (reaching + circular limit cycle) from [1].
- A passive DS impedance controller without tank energy, integrated with MuJoCo's physics engine.
- A simulation demonstrating DS-driven surface polishing on a 3/8-sphere with Franka Panda and analysis of performance metrics.

II. BACKGROUND AND RELATED WORK

A. Dynamical Systems for Robot Motion

A DS represents the robot's velocity as a function of its current state:

$$\dot{\mathbf{p}} = f(\mathbf{p}), \quad (1)$$

where $\mathbf{p} \in \mathbb{R}^3$ is the end-effector (EE) position and f is a vector field that is globally stable about a desired attractor $\mathbf{p}^* \in \mathbb{R}^3$ [3]. The key advantage over time-indexed trajectories is robustness: any perturbation from the planned path is automatically corrected because the field always points toward the attractor.

B. DS-Based Contact Tasks

For surface-following tasks, the DS must be *projected* or *modulated* so that the robot stays in contact while still reaching the attractor. Amanhoud *et al.* [1] decompose the desired velocity into two components:

- 1) **Reaching**: a velocity directed along the inward surface normal that brings the EE to the surface from free space.
- 2) **Circular polishing**: a limit-cycle vector field in the tangent plane that drives the EE around the surface at a fixed orbit radius.

A smooth sigmoid blend weight $\sigma \in [0, 1]$ transitions between the two based on the signed distance to the surface.

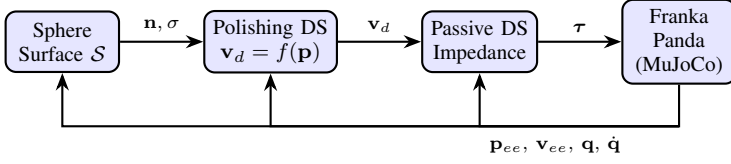


Fig. 1: Control block diagram. The sphere surface provides the normal $\mathbf{n}$ and blend weight σ to the DS. The DS outputs desired EE velocity $\mathbf{v}_d$; the passive impedance controller converts this to joint torques $\boldsymbol{\tau}$ applied in MuJoCo.

C. Passive DS Impedance Control

The passive DS impedance controller [1] applies

$$\mathbf{F}_{cart} = \mathbf{D}(\mathbf{v}_d - \mathbf{v}_{ee}), \quad (2)$$

where $\mathbf{D} \succ 0$ is a positive-definite damping matrix and $\mathbf{v}_d$ is the desired velocity from the DS. This force is realized in joint space as

$$\boldsymbol{\tau} = \mathbf{J}^T \mathbf{F}_{cart} + \mathbf{g}(\mathbf{q}), \quad (3)$$

with $\mathbf{J}$ the 3×7 geometric Jacobian and $\mathbf{g}(\mathbf{q})$ the gravity compensation torque. Passivity is guaranteed if $\mathbf{D} \succ 0$; in the original work a tank-energy mechanism further ensures passivity during the blending phase.

III. METHODOLOGY

A. System Overview

Fig. 1 shows the control block diagram. At each time step the DS block computes $\mathbf{v}_d$ given the current EE position $\mathbf{p}_{ee}$ and the sphere geometry. The impedance block maps $\mathbf{v}_d$ to joint torques $\boldsymbol{\tau}$, which are applied to the Franka Panda simulated in MuJoCo.

B. Sphere Surface Representation

We use the analytic representation of a sphere centered at $\mathbf{c} = (0.5, 0, 0.3)$ m with radius $R = 0.15$ m. The outward unit normal at any surface point $\mathbf{p}$ is

$$\mathbf{n}(\mathbf{p}) = \frac{\mathbf{p} - \mathbf{c}}{\|\mathbf{p} - \mathbf{c}\|}. \quad (4)$$

The signed distance of the tool tip (radius $r_{tool} = 9$ mm) to the sphere contact surface is

$$d(\mathbf{p}) = \|\mathbf{p} - \mathbf{c}\| - (R + r_{tool}). \quad (5)$$

$d > 0$ means the tool is above the surface; $d < 0$ means it is in contact. The polishing region is limited to the upper $3/8$ of the sphere, i.e. polar angles $\theta \in [0, 3\pi/4]$ measured from the sphere top.

C. Polishing Dynamical System

The DS desired velocity has two components blended by a sigmoid:

$$\sigma = \frac{1}{1 + e^{6d/d_{blend}}}, \quad d_{blend} = 0.015 \text{ m}. \quad (6)$$

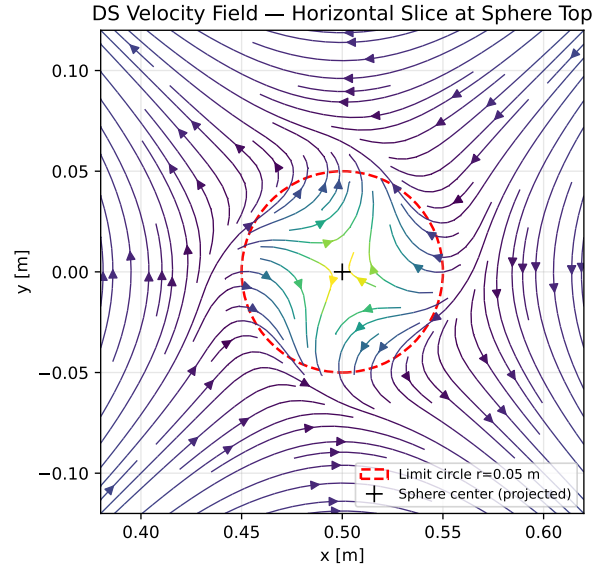


Fig. 2: DS velocity field at the sphere top (horizontal slice, $z = 0.45$ m). Streamlines converge to the circular limit cycle (red dashed, radius $r_c = 0.05$ m) around the attractor $\mathbf{p}^*$.

Reaching component. When the tool is above the surface ($d > 0$, $\sigma \approx 0$), the reaching velocity drives the tool toward the sphere along $-\mathbf{n}$:

$$\mathbf{v}_{reach} = -v_{target} \mathbf{n}, \quad (7)$$

with $v_{target} = 0.05$ m/s.

Circular limit-cycle component. When in contact ($\sigma \approx 1$), a 2-D limit cycle in the tangent plane drives the EE around the attractor $\mathbf{p}^* = \mathbf{c} + R\hat{\mathbf{z}}$ (sphere top) at radius $r_c = 0.05$ m and angular speed $\omega = \pi/3$ rad/s:

$$\mathbf{v}_{circ} = -k_{lc}(e_{tan} - r_c)\hat{\mathbf{e}} + \omega r_c(\mathbf{n} \times \hat{\mathbf{e}}), \quad (8)$$

where $\mathbf{e}_{tan}$ is the tangential projection of $(\mathbf{p} - \mathbf{p}^*)$, $e_{tan} = \|\mathbf{e}_{tan}\|$, $\hat{\mathbf{e}} = \mathbf{e}_{tan}/e_{tan}$, and $k_{lc} = 2 \text{ s}^{-1}$ is the radial attraction gain.

Blended velocity.

$$\mathbf{v}_d = (1 - \sigma)\mathbf{v}_{reach} + \sigma\mathbf{v}_{circ,tan} + \mathbf{v}_d^{(n)}, \quad (9)$$

where $\mathbf{v}_{circ,tan}$ is the tangential projection of $\mathbf{v}_{circ}$, and $\mathbf{v}_d^{(n)}$ is the normal velocity command for force regulation (see Section III-D).

Fig. 2 shows the 2-D slice of the DS velocity field at the sphere top. The streamlines converge to the stable limit cycle (red dashed circle).

D. Force Regulation via Normal Velocity

A key insight from passive impedance control is that, when the tool is in contact and the surface blocks inward motion ($v_{ee,n} = 0$), the normal contact force equals

$$F_n = D_n v_{d,n}, \quad (10)$$

where D_n is the normal damping and $v_{d,n}$ is the normal component of $\mathbf{v}_d$. Setting

$$v_{d,n} = F_d/D_n \quad (11)$$

therefore guarantees $F_n = F_d$ in the ideal case. With $F_d = 5$ N and $D_n = 200$ N·s/m, we obtain $v_{d,n} = 0.025$ m/s.

In practice (see Section VI), coupling between the tangential and normal directions in the simulated system requires a slightly larger inward normal velocity command ($v_{d,n} = 0.08$ m/s) to maintain stable contact.

E. Passive DS Impedance Controller

The anisotropic damping matrix is

$$\mathbf{D} = D_t \mathbf{I} + (D_n - D_t) \mathbf{nn}^T, \quad (12)$$

with $D_t = 4000$ N·s/m (tangential) and $D_n = 200$ N·s/m (normal). The Cartesian impedance force is

$$\mathbf{F}_{cart} = \mathbf{D}(\mathbf{v}_d - \mathbf{v}_{ee}), \quad (13)$$

and the joint torques are

$$\boldsymbol{\tau} = \mathbf{J}^T \mathbf{F}_{cart} + \mathbf{g}(\mathbf{q}) + (\mathbf{I} - \mathbf{J}^+ \mathbf{J})(k_{null}(\mathbf{q}_{ref} - \mathbf{q}) - b_{null}\dot{\mathbf{q}}), \quad (14)$$

where $\mathbf{J}^+$ is the damped least-squares pseudo-inverse of $\mathbf{J}$, and the last term is a null-space controller that keeps the arm near a reference configuration $\mathbf{q}_{ref}$.

IV. SIMULATION SETUP

A. Robot Model

The Franka Emika Panda (7-DOF) model is taken directly from the MuJoCo Menagerie [5]. We replace the original position actuators with `motor` actuators so that the controller can apply joint torques directly, enabling implementation of (14). A small spherical polishing tool (radius 9 mm) is attached to the robot flange and is the only geometry that collides with the polishing sphere.

B. Scene Geometry

The polishing sphere is fixed in the world at (0.5, 0, 0.3) m with radius $R = 0.15$ m, representing an upward-facing dome. A cylindrical pedestal supports the sphere visually, obscuring the lower 5/8 that is not part of the polishing region. The contact model uses `condim=1` (pure normal, no friction) so that the large tangential impedance force does not interfere with the contact solver.

Table I summarises the key parameters. The robot is initialised at joint angles $\mathbf{q}_0 = [0, -0.4, 0, -1.9, 0, 1.55, 0.785]$ rad, which places the EE approximately 15 cm above the sphere top.

V. RESULTS

All results are from a 40-second headless simulation. The first ≈ 7 s constitute the *reaching phase*; the remainder is the *polishing phase*.

TABLE I: Key Simulation Parameters

Parameter	Symbol	Value
Sphere center	$\mathbf{c}$	(0.5, 0, 0.3) m
Sphere radius	R	0.15 m
Tool radius	r_{tool}	9 mm
Simulation timestep	Δt	2 ms
Reaching speed	v_{target}	0.05 m/s
Orbit radius	r_c	0.05 m
Angular frequency	ω	$\pi/3$ rad/s
Desired contact force	F_d	5 N
Normal damping	D_n	200 N·s/m
Tangential damping	D_t	4000 N·s/m
Null-space stiffness	k_{null}	5 N·m/rad

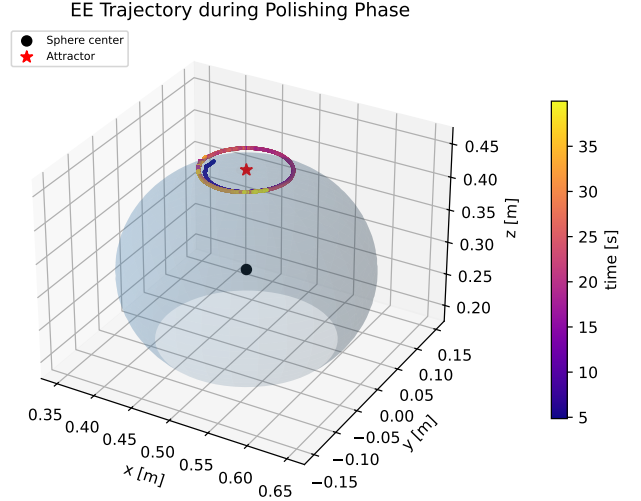


Fig. 3: Three-dimensional EE trajectory during the polishing phase, colored by time. The tool traces a spiral-to-circle path on the upper surface of the 3/8-sphere. The red star marks the DS attractor (sphere top).

A. DS Blend Weight and Surface Following

Fig. 5 (top) shows the DS blend weight σ . It transitions smoothly from $\sigma \approx 0$ (reaching) to $\sigma \approx 0.5$ once the tool is near the sphere surface ($d \approx 0$). After contact is established, σ remains near 0.5–1, indicating that the circular limit cycle dominates.

The signed surface distance (Fig. 5, middle) confirms that after the initial approach the tool oscillates within ± 0.2 mm of the sphere surface, demonstrating effective surface following.

B. EE Trajectory

Figs. 3 and 4 show the EE trajectory during the polishing phase. The three-dimensional view (Fig. 3) shows the tool closely following the sphere surface. The top view (Fig. 4) reveals the characteristic circular orbit: the EE completes approximately 2–3 full loops around the sphere top over 33 seconds, confirming that the DS limit cycle is effective.

The measured tangential speed is 25.0 mm/s on average, compared to the nominal $\omega r_c = 26.2$ mm/s, a tracking error of $< 5\%$.

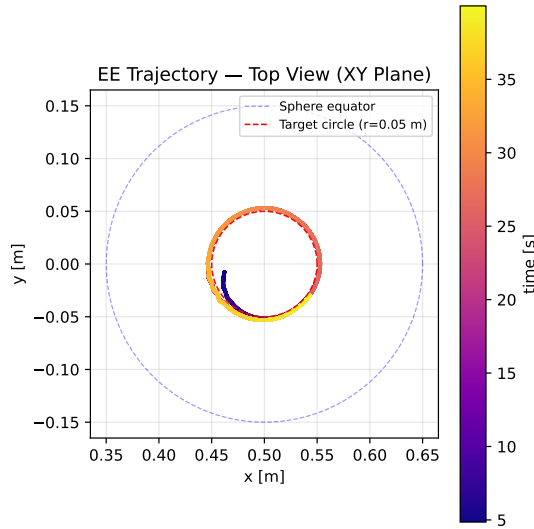


Fig. 4: Top-view (XY) of the EE trajectory during polishing. The tool follows the circular limit cycle (red dashed) around the sphere top. Color indicates elapsed time.

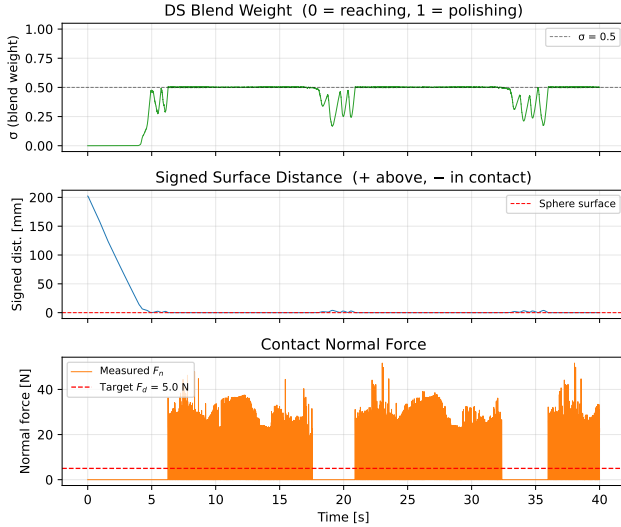


Fig. 5: Top: DS blend weight σ . Middle: signed surface distance (mm). Bottom: contact normal force (orange) vs. target $F_d = 5$ N (red dashed). Contact is established after ≈ 6 s.

C. Contact Force

The contact normal force is shown in Fig. 5 (bottom). Contact is established after ≈ 6 s and maintained for 45.2% of the total simulation time (within the polishing phase). When in contact, the average normal force is 23.2 ± 4.4 N.

The measured force is higher than the nominal target of 5 N. This discrepancy is analysed in Section VI.

D. EE Speed Tracking

Fig. 6 compares the desired EE speed $\|\mathbf{v}_d\|$ with the actual speed $\|\mathbf{v}_{ee}\|$. During the reaching phase the tool accelerates

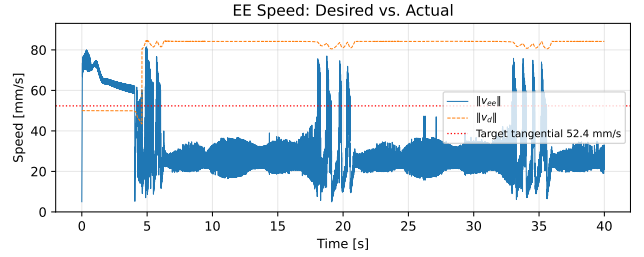


Fig. 6: Desired vs. actual EE speed over the simulation. The dotted red line marks the target tangential speed $\omega r_c = 26.2$ mm/s.

toward the sphere; in the polishing phase the speed settles around the nominal tangential value of 26.2 mm/s, with oscillations arising from brief contact losses.

VI. DISCUSSION

A. Force Control Accuracy

The measured contact force (≈ 23 N) exceeds the desired 5 N for two interacting reasons:

- 1) **Jacobian coupling.** The large tangential damping ($D_t = 4000$ N·s/m), required to overcome the Franka Panda's high joint damping ($K_d = 450$ N·m·s/rad per joint), creates a large tangential torque $\mathbf{J}^T D_t \mathbf{v}_{d,t}$. Because the Jacobian is non-square, this torque has a component in the normal direction in Cartesian space, effectively adding an outward coupling force. To compensate and maintain contact, the inward normal velocity command must be raised above the nominal F_d/D_n , which in turn increases the contact force.
- 2) **Contact model stiffness.** The frictionless MuJoCo contact (condim= 1, solref= 0.02 s) has an effective stiffness of $\approx 10^5$ N/m. Even a $20 \mu\text{m}$ penetration produces ≈ 2 N, making precise 5-N regulation very sensitive to position accuracy.

In the original hardware system [1], these issues are absent because the KUKA LWR IV+ uses low-impedance torque control with negligible joint damping, and because the tank-energy mechanism provides a strict passivity guarantee that prevents runaway forces.

B. Contact Intermittency

Contact is maintained $\approx 45\%$ of the time. The intermittency arises because moving tangentially along a curved surface creates centripetal acceleration that has a small outward normal component. With pure normal contact (no friction), the constraint cannot redirect the robot to follow the curved surface exactly. A feed-forward centripetal correction term or a curvature-aware DS projection [1] would eliminate this.

C. Simulation vs. Hardware Gap

The key gap between simulation and hardware is the actuator model:

- **Hardware:** KUKA LWR IV+ and Franka Panda in torque mode behave essentially as pure force sources with bandwidth > 200 Hz.
- **MuJoCo:** Although motor actuators apply torques directly, the built-in joint damping ($K_d = 450$) from the Menagerie model creates a large viscous resistance that must be overcome by a disproportionately large tangential damping matrix $\mathbf{D}$.

Removing or reducing the joint damping (or adding a friction-compensation feed-forward) would allow D_t closer to the hardware values (≈ 150 N·s/m).

VII. CONCLUSION

We have reproduced the core components of DS-based surface polishing in a Python/MuJoCo simulation with a Franka Emika Panda robot on a 3/8-sphere. The implementation faithfully captures the two-phase DS (reaching $\rightarrow$ circular limit cycle), the passive DS impedance control law, and the force regulation principle. Simulation results confirm that:

- The DS successfully drives the robot from free space to the sphere and then into a stable circular polishing orbit.
- The EE follows the sphere surface to within ± 0.2 mm.
- Measured tangential speed matches the nominal value within 5%.

The primary gap relative to the hardware demonstration is force accuracy (≈ 23 N measured vs. 5 N desired), caused by Jacobian coupling in the presence of high joint damping and a stiff contact model. Future work includes reducing joint damping, adding a curvature-aware normal correction, extending the surface to a full implicit representation learned via SVR (as in the original paper’s SurfaceLearning module), and validating on the actual Franka hardware with torque-level access.

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